Econometrics

Tidying 🧹, Visualising 📈 and Summarising 📊 Data

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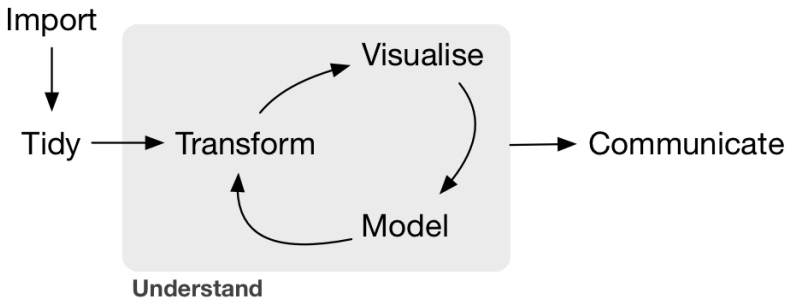
Working With Data

- Econometrics is about `data`.



Working With Data

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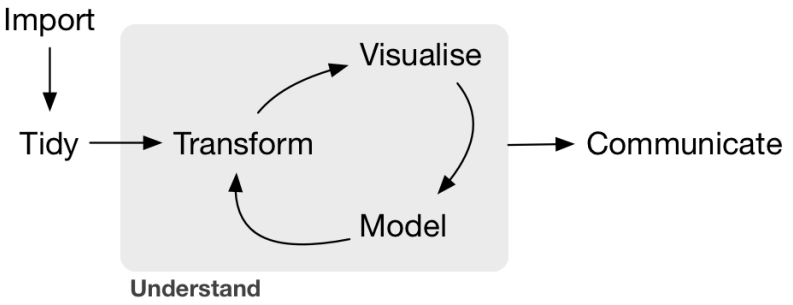

- According a to 2014 NYTimes article, "data scientists [...] spend from ***50 percent to 80 percent of their time*** mired in this more mundane labor of collecting and preparing unruly digital data, before it can be explored for useful nuggets."

- In the next two lectures you will learn the basics of ***tidying***, ***visualising*** and ***summarising*** data

Tidying Data 🧹

Intro to `dplyr`

- `dplyr` is part of the `tidyverse` package family.
- `data.table` is an alternative. Very fast and highly recommended for **big** data. Look here for an intro!
- Both have pros and cons. We'll start you off with `dplyr`.

`dplyr` Overview

- You are ***highly encouraged*** to read through Hadley Wickham's chapter. It's clear and concise.

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- We operate on `data.frames` or `tibbles` (*nicer looking* data.frames.)

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$$\text{verb}(\underbrace{\text{data.frame}}_{\text{1st argument}}, \underbrace{\text{what to do}}_{\text{2nd argument}})$$

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- We operate on `data.frames` or `tibbles` (*nicer looking* data.frames.)
- All *verbs* work as follows:

$$\text{verb}(\underbrace{\text{data.frame}}_{\text{1st argument}}, \underbrace{\text{what to do}}_{\text{2nd argument}})$$

- Alternatively you can (should) use the `pipe` operator `%>%` (part of `magrittr` package):

$$\underbrace{\text{data.frame}}_{\text{1st argument}} \quad \underbrace{\%>\%}_{\text{"pipe" operator}} \quad \text{verb}(\underbrace{\text{what to do}}_{\text{2nd argument}})$$

Main `dplyr` Verbs

1. `filter()`: Choose observations based on a certain condition (i.e. subset)

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3. `select()`: Select variables by name
4. `mutate()`: Create new variables out of existing ones
5. `summarise()`: Collapse data to a single summary
6. `group_by()`: All the above can be used in conjunction with `group_by()` to use functions on groups rather than entire data

Data: 2016 US election polls from the `dslabs` package

- This dataset contains **real** data on polls made during the 2016 US Presidential elections and compiled by fivethirtyeight

```r
library(dslabs)
library(tidyverse)
data(polls_us_election_2016, package = "dslabs") # load data from package
polls_us_election_2016 <- as_tibble(polls_us_election_2016) # convert to a tibble
head(polls_us_election_2016[,1:6]) # show first 6 lines of first 6 variables
```

```
## # A tibble: 6 × 6
##   state startdate  enddate    pollster                          grade samplesize
##   <fct> <date>     <date>     <fct>                             <fct>      <int>
## 1 U.S.  2016-11-03 2016-11-06 ABC News/Washington Post          A+          2220
## 2 U.S.  2016-11-01 2016-11-07 Google Consumer Surveys           B          26574
## 3 U.S.  2016-11-02 2016-11-06 Ipsos                             A-          2195
## 4 U.S.  2016-11-04 2016-11-07 YouGov                            B           3677
## 5 U.S.  2016-11-03 2016-11-06 Gravis Marketing                  B-         16639
## 6 U.S.  2016-11-03 2016-11-06 Fox News/Anderson Robbins Resear… A           1295
```

🚨 This is a `tibble` (more informative than `data.frame`)

Data: 2016 US election polls from the `dslabs` package

What variables does this dataset contain?

```
str(polls_us_election_2016)
```

```
## tibble [4,208 × 15] (S3: tbl_df/tbl/data.frame)
##  $ state           : Factor w/ 57 levels "Alabama","Alaska",..: 50 50 50 50 50 50 50 50 37 50 ...
##  $ startdate       : Date[1:4208], format: "2016-11-03" "2016-11-01" ...
##  $ enddate         : Date[1:4208], format: "2016-11-06" "2016-11-07" ...
##  $ pollster        : Factor w/ 196 levels "ABC News/Washington Post",..: 1 63 81 194 65 55 18 113 195 76 ...
##  $ grade           : Factor w/ 10 levels "D","C-","C","C+",..: 10 6 8 6 5 9 8 8 NA 8 ...
##  $ samplesize      : int [1:4208] 2220 26574 2195 3677 16639 1295 1426 1282 8439 1107 ...
##  $ population      : chr [1:4208] "lv" "lv" "lv" "lv" ...
##  $ rawpoll_clinton : num [1:4208] 47 38 42 45 47 ...
##  $ rawpoll_trump   : num [1:4208] 43 35.7 39 41 43 ...
##  $ rawpoll_johnson : num [1:4208] 4 5.46 6 5 3 3 5 6 6 7.1 ...
##  $ rawpoll_mcmullin: num [1:4208] NA NA NA NA NA NA NA NA NA NA ...
##  $ adjpoll_clinton : num [1:4208] 45.2 43.3 42 45.7 46.8 ...
##  $ adjpoll_trump   : num [1:4208] 41.7 41.2 38.8 40.9 42.3 ...
##  $ adjpoll_johnson : num [1:4208] 4.63 5.18 6.84 6.07 3.73 ...
##  $ adjpoll_mcmullin: num [1:4208] NA NA NA NA NA NA NA NA NA NA ...
```

`dplyr` Verbs

Filter observations

```
filter()
```

Example: Which polls had a sample size of at least 2,000 people?

`dplyr` Verbs

Filter observations

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Example: Which polls had a sample size of at least 2,000 people?

```
polls_us_election_2016
```

dplyr Verbs

Filter observations

```
filter()
```

Example: Which polls had a sample size of at least 2,000 people?

```
polls_us_election_2016 %>%
  filter(samplesize > 2000)
```

```
## # A tibble: 403 × 15
##    state      startdate  enddate    pollster         grade samplesize population
##    <fct>      <date>     <date>     <fct>            <fct>      <int> <chr>
##  1 U.S.       2016-11-03 2016-11-06 ABC News/Washin… A+          2220 lv
##  2 U.S.       2016-11-01 2016-11-07 Google Consumer… B          26574 lv
##  3 U.S.       2016-11-02 2016-11-06 Ipsos            A-          2195 lv
##  4 U.S.       2016-11-04 2016-11-07 YouGov           B           3677 lv
##  5 U.S.       2016-11-03 2016-11-06 Gravis Marketing B-         16639 rv
##  6 New Mexico 2016-11-06 2016-11-06 Zia Poll         <NA>        8439 lv
##  7 U.S.       2016-11-05 2016-11-07 The Times-Picay… <NA>        2521 lv
##  8 U.S.       2016-11-01 2016-11-07 USC Dornsife/LA… <NA>        2972 lv
##  9 Georgia    2016-11-03 2016-11-06 Gravis Marketing B-          2002 rv
## 10 Virginia   2016-11-01 2016-11-02 Remington        <NA>        3076 lv
## # ℹ 393 more rows
## # ℹ 8 more variables: rawpoll_clinton <dbl>, rawpoll_trump <dbl>,
## #   rawpoll_johnson <dbl>, rawpoll_mcmullin <dbl>, adjpoll_clinton <dbl>,
## #   adjpoll_trump <dbl>, adjpoll_johnson <dbl>, adjpoll_mcmullin <dbl>
```

`dplyr` Verbs

Filter observations

```
filter()
```

Standard suite of comparison operators:

- `>`: greater than,
- `<`: smaller than,
- `>=`: greater than or equal to,
- `<=`: smaller than or equal to,
- `!=`: not equal to,
- `==`: equal to.

Logical operators:

1. `x & y`: `x` **and** `y`
2. `x | y`: `x` **or** `y`
3. `!y`: **not** `y`

`dplyr` Verbs

Filter observations

```
filter()
```

`R` has the convenient `x %in% y` operator (conversely `!(x %in% y)`), `TRUE` if `x` is *a member of* `y`.

```
3 %in% 1:3
```

```
## [1] TRUE
```

```
c(2,5) %in% 2:10  # also vectorized
```

```
## [1] TRUE TRUE
```

```
c("S","Po") %in% c("Sciences","Po")  # also strings
```

```
## [1] FALSE  TRUE
```

`dplyr` Verbs

Filter observations

```
filter()
```

Example: Which A graded poll with at least 2,000 people had Trump win at least 45% of the vote?

`dplyr` Verbs

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```
polls_us_election_2016
```

`dplyr` Verbs

Filter observations

```
filter()
```

Example: Which A graded poll with at least 2,000 people had Trump win at least 45% of the vote?

```
polls_us_election_2016 %>%
  filter(grade == "A" & samplesize > 2000 & rawpoll_trump > 45)
```

```
## # A tibble: 1 × 15
##   state   startdate  enddate    pollster       grade samplesize population
##   <fct>   <date>     <date>     <fct>          <fct>      <int> <chr>
## 1 Indiana 2016-04-26 2016-04-28 Marist College A           2149 rv
## # ℹ 8 more variables: rawpoll_clinton <dbl>, rawpoll_trump <dbl>,
## #   rawpoll_johnson <dbl>, rawpoll_mcmullin <dbl>, adjpoll_clinton <dbl>,
## #   adjpoll_trump <dbl>, adjpoll_johnson <dbl>, adjpoll_mcmullin <dbl>
```

`dplyr` Verbs

Filter

Create new variable(s)

```
mutate()
```

Example: What was

1. the combined vote share of Trump and Clinton for each poll?
2. the difference between Trump's raw poll vote share and 538's adjusted vote share?

`dplyr` Verbs

Filter

Create new variable(s)

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```
polls_us_election_2016
```

`dplyr` Verbs

Filter

Create new variable(s)

```
mutate()
```

Example: What was

1. the combined vote share of Trump and Clinton for each poll?
2. the difference between Trump's raw poll vote share and 538's adjusted vote share?

```
polls_us_election_2016 %>%
  mutate(trump_clinton_tot = rawpoll_trump + rawpoll_clinton,
         trump_raw_adj_diff = rawpoll_trump - adjpoll_trump)
```

```
## # A tibble: 4,208 × 17
##    state      startdate  enddate    pollster         grade samplesize population
##    <fct>      <date>     <date>     <fct>            <fct>      <int> <chr>
##  1 U.S.       2016-11-03 2016-11-06 ABC News/Washin… A+          2220 lv
##  2 U.S.       2016-11-01 2016-11-07 Google Consumer… B          26574 lv
##  3 U.S.       2016-11-02 2016-11-06 Ipsos            A-          2195 lv
##  4 U.S.       2016-11-04 2016-11-07 YouGov           B           3677 lv
##  5 U.S.       2016-11-03 2016-11-06 Gravis Marketing B-         16639 rv
##  6 U.S.       2016-11-03 2016-11-06 Fox News/Anders… A           1295 lv
##  7 U.S.       2016-11-02 2016-11-06 CBS News/New Yo… A-          1426 lv
##  8 U.S.       2016-11-03 2016-11-05 NBC News/Wall S… A-          1282 lv
##  9 New Mexico 2016-11-06 2016-11-06 Zia Poll         <NA>        8439 lv
## 10 U.S.       2016-11-04 2016-11-07 IBD/TIPP         A-          1107 lv
## # ℹ 4,198 more rows
## # ℹ 10 more variables: rawpoll_clinton <dbl>, rawpoll_trump <dbl>,
```

`dplyr` Verbs

Filter

Create new variable(s)

```
mutate()
```

Example: What was

1. the combined vote share of Trump and Clinton for each poll?
2. the difference between Trump's raw poll vote share and 538's adjusted vote share?

```
polls_us_election_2016 %>%
  mutate(trump_clinton_tot = rawpoll_trump + rawpoll_clinton,
         trump_raw_adj_diff = rawpoll_trump - adjpoll_trump) %>%
  names()
```

```
##  [1] "state"              "startdate"          "enddate"
##  [4] "pollster"           "grade"              "samplesize"
##  [7] "population"         "rawpoll_clinton"    "rawpoll_trump"
## [10] "rawpoll_johnson"    "rawpoll_mcmullin"   "adjpoll_clinton"
## [13] "adjpoll_trump"      "adjpoll_johnson"    "adjpoll_mcmullin"
## [16] "trump_clinton_tot"  "trump_raw_adj_diff"
```

`dplyr` Verbs

Filter

Mutate

Keep some variable(s)

```
select()
```

Example: Only keep the variables `state, startdate, enddate, pollster, rawpoll_clinton, rawpoll_trump`

`dplyr` Verbs

Filter

Mutate

Keep some variable(s)

```
select()
```

Example: Only keep the variables `state, startdate, enddate, pollster, rawpoll_clinton, rawpoll_trump`

```
polls_us_election_2016
```

```
## # A tibble: 10 × 15
##    state      startdate  enddate    pollster         grade samplesize population
##    <fct>      <date>     <date>     <fct>            <fct>      <int> <chr>
##  1 U.S.       2016-11-03 2016-11-06 ABC News/Washin… A+          2220 lv
##  2 U.S.       2016-11-01 2016-11-07 Google Consumer… B          26574 lv
##  3 U.S.       2016-11-02 2016-11-06 Ipsos            A-          2195 lv
##  4 U.S.       2016-11-04 2016-11-07 YouGov           B           3677 lv
##  5 U.S.       2016-11-03 2016-11-06 Gravis Marketing B-         16639 rv
##  6 U.S.       2016-11-03 2016-11-06 Fox News/Anders… A           1295 lv
##  7 U.S.       2016-11-02 2016-11-06 CBS News/New Yo… A-          1426 lv
##  8 U.S.       2016-11-03 2016-11-05 NBC News/Wall S… A-          1282 lv
##  9 New Mexico 2016-11-06 2016-11-06 Zia Poll         <NA>        8439 lv
## 10 U.S.       2016-11-04 2016-11-07 IBD/TIPP         A-          1107 lv
## # ℹ 8 more variables: rawpoll_clinton <dbl>, rawpoll_trump <dbl>,
## #   rawpoll_johnson <dbl>, rawpoll_mcmullin <dbl>, adjpoll_clinton <dbl>,
## #   adjpoll_trump <dbl>, adjpoll_johnson <dbl>, adjpoll_mcmullin <dbl>
```

`dplyr` Verbs

Filter

Mutate

Keep some variable(s)

```
select()
```

Example: Only keep the variables `state, startdate, enddate, pollster, rawpoll_clinton, rawpoll_trump`

```
polls_us_election_2016 %>%
  select(state,startdate,enddate,pollster,rawpoll_clinton,rawpoll_trump)
```

```
## # A tibble: 4,208 × 6
##    state      startdate  enddate    pollster       rawpoll_clinton rawpoll_trump
##    <fct>      <date>     <date>     <fct>                    <dbl>         <dbl>
##  1 U.S.       2016-11-03 2016-11-06 ABC News/Wash…            47            43
##  2 U.S.       2016-11-01 2016-11-07 Google Consum…            38.0          35.7
##  3 U.S.       2016-11-02 2016-11-06 Ipsos                     42            39
##  4 U.S.       2016-11-04 2016-11-07 YouGov                    45            41
##  5 U.S.       2016-11-03 2016-11-06 Gravis Market…            47            43
##  6 U.S.       2016-11-03 2016-11-06 Fox News/Ande…            48            44
##  7 U.S.       2016-11-02 2016-11-06 CBS News/New …            45            41
##  8 U.S.       2016-11-03 2016-11-05 NBC News/Wall…            44            40
##  9 New Mexico 2016-11-06 2016-11-06 Zia Poll                  46            44
## 10 U.S.       2016-11-04 2016-11-07 IBD/TIPP                  41.2          42.7
## # ℹ 4,198 more rows
```

`dplyr` Verbs

Filter

Mutate

Keep some variable(s)

```
select()
```

Example: Only keep the variables `state, startdate, enddate, pollster, rawpoll_clinton, rawpoll_trump`

```
polls_us_election_2016 %>%
  select(state,startdate,enddate,pollster,rawpoll_clinton,rawpoll_trump) %>%
  names()
```

```
## [1] "state"           "startdate"       "enddate"         "pollster"
## [5] "rawpoll_clinton" "rawpoll_trump"
```

`dplyr` Verbs

Filter

Mutate

Select

Compute statistic

```
summarise()
```

Example: What is the maximum vote share for Trump?

`dplyr` Verbs

Filter

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Example: What is the maximum vote share for Trump?

```
polls_us_election_2016
```

`dplyr` Verbs

Filter

Mutate

Select

Compute statistic

```
summarise()
```

Example: What is the maximum vote share for Trump?

```
polls_us_election_2016 %>%
  summarise(max_trump = max(rawpoll_trump))
```

```
## # A tibble: 1 × 1
##   max_trump
##       <dbl>
## 1        68
```

`dplyr` Verbs

Filter

Mutate

Select

Summarise

Apply function by group

```
group_by()
```

Example: What is the average vote share for Clinton by poll grade?

`dplyr` Verbs

Filter

Mutate

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Apply function by group

```
group_by()
```

Example: What is the average vote share for Clinton by poll grade?

```
polls_us_election_2016
```

`dplyr` Verbs

Filter

Mutate

Select

Summarise

Apply function by group

```
group_by()
```

Example: What is the average vote share for Clinton by poll grade?

```
polls_us_election_2016 %>%
  group_by(grade)
```

```
## # A tibble: 4,208 × 15
## # Groups:   grade [11]
##    state      startdate  enddate    pollster         grade samplesize population
##    <fct>      <date>     <date>     <fct>            <fct>      <int> <chr>
##  1 U.S.       2016-11-03 2016-11-06 ABC News/Washin… A+          2220 lv
##  2 U.S.       2016-11-01 2016-11-07 Google Consumer… B          26574 lv
##  3 U.S.       2016-11-02 2016-11-06 Ipsos            A-          2195 lv
##  4 U.S.       2016-11-04 2016-11-07 YouGov           B           3677 lv
##  5 U.S.       2016-11-03 2016-11-06 Gravis Marketing B-         16639 rv
##  6 U.S.       2016-11-03 2016-11-06 Fox News/Anders… A           1295 lv
##  7 U.S.       2016-11-02 2016-11-06 CBS News/New Yo… A-          1426 lv
##  8 U.S.       2016-11-03 2016-11-05 NBC News/Wall S… A-          1282 lv
##  9 New Mexico 2016-11-06 2016-11-06 Zia Poll         <NA>        8439 lv
## 10 U.S.       2016-11-04 2016-11-07 IBD/TIPP         A-          1107 lv
## # ℹ 4,198 more rows
## # ℹ 8 more variables: rawpoll_clinton <dbl>, rawpoll_trump <dbl>,
## #   rawpoll_johnson <dbl>, rawpoll_mcmullin <dbl>, adjpoll_clinton <dbl>,
## #   adjpoll_trump <dbl>, adjpoll_johnson <dbl>, adjpoll_mcmullin <dbl>
```

`dplyr` Verbs

Filter

Mutate

Select

Summarise

Apply function by group

```
group_by()
```

Example: What is the average vote share for Clinton by poll grade?

```
polls_us_election_2016 %>%
  group_by(grade) %>%
  class()
```

```
## [1] "grouped_df" "tbl_df"     "tbl"        "data.frame"
```

`dplyr` Verbs

Filter

Mutate

Select

Summarise

Apply function by group

```
group_by()
```

Example: What is the average vote share for Clinton by poll grade?

```
polls_us_election_2016 %>%
  group_by(grade)
```

```
## # A tibble: 4,208 × 15
## # Groups:   grade [11]
##    state      startdate  enddate    pollster         grade samplesize population
##    <fct>      <date>     <date>     <fct>            <fct>      <int> <chr>
##  1 U.S.       2016-11-03 2016-11-06 ABC News/Washin… A+          2220 lv
##  2 U.S.       2016-11-01 2016-11-07 Google Consumer… B          26574 lv
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##  8 U.S.       2016-11-03 2016-11-05 NBC News/Wall S… A-          1282 lv
##  9 New Mexico 2016-11-06 2016-11-06 Zia Poll         <NA>        8439 lv
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## # ℹ 4,198 more rows
## # ℹ 8 more variables: rawpoll_clinton <dbl>, rawpoll_trump <dbl>,
## #   rawpoll_johnson <dbl>, rawpoll_mcmullin <dbl>, adjpoll_clinton <dbl>,
## #   adjpoll_trump <dbl>, adjpoll_johnson <dbl>, adjpoll_mcmullin <dbl>
```

`dplyr` Verbs

Filter

Mutate

Select

Summarise

Apply function by group

```
group_by()
```

Example: What is the average vote share for Clinton by poll grade?

```
polls_us_election_2016 %>%
  group_by(grade) %>%
  summarise(mean_vote_clinton = mean(rawpoll_clinton))
```

```
## # A tibble: 11 × 2
##    grade mean_vote_clinton
##    <fct>             <dbl>
##  1 D                  46.7
##  2 C-                 43.2
##  3 C                  41.8
##  4 C+                 44.2
##  5 B-                 43.9
##  6 B                  37.3
##  7 B+                 44.1
##  8 A-                 43.0
##  9 A                  45.3
## 10 A+                 45.8
## 11 <NA>               43.2
```

Chaining ⛓ Commands Together

Works for all `dplyr` verbs:

```
polls_us_election_2016
```

Chaining ⛓ Commands Together

Works for all `dplyr` verbs:

```
polls_us_election_2016 %>%
    mutate(trump_clinton_diff =
             rawpoll_trump -
             rawpoll_clinton)
```

```
## # A tibble: 4,208 × 16
##    state      startdate  enddate    pollster          grade samplesize pop
##    <fct>      <date>     <date>     <fct>             <fct>      <int> <ch
##  1 U.S.       2016-11-03 2016-11-06 ABC News/Washin… A+          2220 lv
##  2 U.S.       2016-11-01 2016-11-07 Google Consumer… B          26574 lv
##  3 U.S.       2016-11-02 2016-11-06 Ipsos            A-          2195 lv
##  4 U.S.       2016-11-04 2016-11-07 YouGov           B           3677 lv
##  5 U.S.       2016-11-03 2016-11-06 Gravis Marketing B-         16639 rv
##  6 U.S.       2016-11-03 2016-11-06 Fox News/Anders… A           1295 lv
##  7 U.S.       2016-11-02 2016-11-06 CBS News/New Yo… A-          1426 lv
##  8 U.S.       2016-11-03 2016-11-05 NBC News/Wall S… A-          1282 lv
##  9 New Mexico 2016-11-06 2016-11-06 Zia Poll         <NA>        8439 lv
## 10 U.S.       2016-11-04 2016-11-07 IBD/TIPP         A-          1107 lv
## # ℹ 4,198 more rows
## # ℹ 9 more variables: rawpoll_clinton <dbl>, rawpoll_trump <dbl>,
## #   rawpoll_johnson <dbl>, rawpoll_mcmullin <dbl>, adjpoll_clinton <dbl>,
## #   adjpoll_trump <dbl>, adjpoll_johnson <dbl>, adjpoll_mcmullin <dbl>,
## #   trump_clinton_diff <dbl>
```

Chaining ⛓ Commands Together

Works for all `dplyr` verbs:

```
polls_us_election_2016 %>%
    mutate(trump_clinton_diff =
             rawpoll_trump -
             rawpoll_clinton) %>%
    filter(trump_clinton_diff>5 &
           state == "Iowa" &
           is.na(rawpoll_johnson))
```

```
## # A tibble: 3 × 16
##   state startdate  enddate    pollster grade samplesize population
##   <fct> <date>     <date>     <fct>    <fct>      <int> <chr>
## 1 Iowa  2016-09-09 2016-09-29 Ipsos    A-           343 lv
## 2 Iowa  2016-09-02 2016-09-22 Ipsos    A-           344 lv
## 3 Iowa  2016-08-26 2016-09-15 Ipsos    A-           347 lv
## # ℹ 9 more variables: rawpoll_clinton <dbl>, rawpoll_trump <dbl>,
## #   rawpoll_johnson <dbl>, rawpoll_mcmullin <dbl>, adjpoll_clinton <dbl>,
## #   adjpoll_trump <dbl>, adjpoll_johnson <dbl>, adjpoll_mcmullin <dbl>,
## #   trump_clinton_diff <dbl>
```

Chaining ⛓ Commands Together

Works for all `dplyr` verbs:

```
polls_us_election_2016 %>%
    mutate(trump_clinton_diff =
             rawpoll_trump -
             rawpoll_clinton) %>%
    filter(trump_clinton_diff>5 &
           state == "Iowa" &
           is.na(rawpoll_johnson)) %>%
    select(pollster)
```

```
## # A tibble: 3 × 1
##   pollster
##   <fct>
## 1 Ipsos
## 2 Ipsos
## 3 Ipsos
```

Chaining ⛓ Commands Together

Works for all `dplyr` verbs:

```
polls_us_election_2016 %>%
    mutate(trump_clinton_diff =
             rawpoll_trump -
             rawpoll_clinton) %>%
    filter(trump_clinton_diff>5 &
           state == "Iowa" &
           is.na(rawpoll_johnson)) %>%
    pull(pollster)
```

```
## [1] Ipsos Ipsos Ipsos
## 196 Levels: ABC News/Washington Post ... Zogby Interactive/JZ Analytics
```

Chaining ⛓ Commands Together

But also with other `R` commands:

```
polls_us_election_2016$samplesize %>% mean(na.rm = T)
```

```
## [1] 1148.216
```

Chaining ⛓️ Commands Together

But also with other `R` commands:

```
polls_us_election_2016$samplesize %>% mean(na.rm = T)
```

```
## [1] 1148.216
```

```
polls_us_election_2016 %>% count()
```

```
## # A tibble: 1 × 1
##       n
##   <int>
## 1  4208
```

(only for 🤓 nerds:)

- The `%>%` pipe is part of the `magrittr` package. `R v4.1.0` adds a *native pipe* via `|>`. you could use it like

```
polls_us_election_2016$samplesize |> mean(na.rm = T)
```

```
## [1] 1148.216
```

Missing Values: `NA`

- Whenever a value is *missing*, we code it as `NA`.

```
x <- NA
```

- `R` propagates `NA` through operations:

```
NA > 5
```

```
## [1] NA
```

```
NA + 10
```

```
## [1] NA
```

- `is.na(x)` function returns `TRUE` if `x` is an `NA`.

```
is.na(x)
```

```
## [1] TRUE
```

Missing Values: `NA`

- Whenever a value is *missing*, we code it as `NA`.

```
x <- NA
```

- `R` propagates `NA` through operations:

```
NA > 5
```

```
## [1] NA
```

```
NA + 10
```

```
## [1] NA
```

- `is.na(x)` function returns `TRUE` if `x` is an `NA`.

```
is.na(x)
```

```
## [1] TRUE
```

- What is confusing is that

```
NA == NA
```

```
## [1] NA
```

- It's easy to illustrate like that:

```
# Let x be Mary's age. We don't know how old she
x <- NA

# Let y be John's age. We don't know how old he i
y <- NA

# Are John and Mary the same age?
x == y
```

```
## [1] NA
```

```
#> [1] NA
# We don't know!
```

Task 1: Data wrangling

10:00

Load the data by running the following code:

```
library(dslabs)
data(polls_us_election_2016)
```

1. Which polls had a missing `grade`?

2. Which polls were (i) polled by American Strategies, GfK Group or Merrill Poll, (ii) had a sample size greater than 1,000, *and* (iii) started on October 20th, 2016? (*Hint: for (i)* `%in%` *might come in handy. Recall that vectors are created using the* `c()` *function. For (iii) make sure to check the format of the variable containing the poll's start date.*)

3. Which polls (i) did not have missing poll data for Johnson, (ii) had a combined raw poll vote share for Trump and Clinton greater than 95% *and* (iii) were done in the state of Ohio? (*Hint: it might be practical to first create a variable containing the combined raw poll vote share for Trump and Clinton and then filter.*)

4. Which state had the highest average Trump vote share for polls which had at least a sample size of 2,000? (*Hint: you'll have to use* `filter`, `group_by`, `summarise` *and* `arrange`. *To obtain ranking in descending order check* `arrange`*'s help page.*)

Visualising Data

Base `R` and `ggplot2`

- Base `R` plotting is fairly good.
- There is an extremely powerful alternative: `ggplot2` (part of the `tidyverse` suite) $\rightarrow$ what we'll be using
- Let's go back to the `gapminder` dataset to run the examples.

The `gapminder` dataset: Overview

- Let's first load the `gapminder` dataset with these commands:

```
library(dslabs)
data(gapminder, package = "dslabs")
```

The `gapminder` dataset: Overview

- Let's first load the `gapminder` dataset with these commands:

```
library(dslabs)
data(gapminder, package = "dslabs")
```

- Here are the first 3 rows and last 2 rows.

```
head(gapminder, n = 3)
```

```
##   country year infant_mortality life_expectancy fertility population
## 1 Albania 1960            115.4           62.87      6.19    1636054
## 2 Algeria 1960            148.2           47.50      7.65   11124892
## 3  Angola 1960            208.0           35.98      7.32    5270844
##           gdp continent          region
## 1          NA    Europe Southern Europe
## 2 13828152297    Africa Northern Africa
## 3          NA    Africa   Middle Africa
```

```
tail(gapminder, n = 2)
```

```
##        country year infant_mortality life_expectancy fertility population gdp
## 10544   Zambia 2016               NA           57.10        NA         NA  NA
## 10545 Zimbabwe 2016               NA           61.69        NA         NA  NA
##       continent         region
## 10544    Africa Eastern Africa
## 10545    Africa Eastern Africa
```

Task 2: Understanding the data

05:00

Load the data by running the following code:

```
library(dslabs)
data(gapminder, package = "dslabs")
```

1. Compute the average population per continent per year, `mean_pop`, and assign the output to a new object `gapminder_mean`. (*Hint: you should have one observation (row) per continent for each year. You'll have to use* `group_by` *and* `summarise`.)

gg is for Grammar of Graphics[1]

[1]: The following slides are taken from Garrick Aden-Buie's wonderful Gentle Guide to the Grammar of Graphics with `ggplot2`

gg is for Grammar of Graphics

Data

```
data %>%
  ggplot()
```

or

```
ggplot(data)
```

gg is for Grammar of Graphics

Data

```
data %>%
  ggplot()
```

or

```
ggplot(data)
```

Tidy Data

1. Each variable forms a ***column***
2. Each observation forms a ***row***
3. Each observational unit forms a table

gg is for Grammar of Graphics

Data

```
data %>%
  ggplot()
```

or

```
ggplot(data)
```

Tidy Data

1. Each variable forms a ***column***
2. Each observation forms a ***row***
3. Each observational unit forms a table

Start by asking

1. What information do I want to use in my visualization?
2. Is that data contained in ***one column/row*** for a given data point?

gg is for Grammar of Graphics

Data

Aesthetics

```
+ aes()
```

Map data to visual elements or parameters

- year
- population
- country

gg is for Grammar of Graphics

Data

Aesthetics

```
+ aes()
```

Map data to visual elements or parameters

- year → **x**
- population → **y**
- country → *shape*, *color*, etc.

gg is for Grammar of Graphics

Data

Aesthetics

```
+ aes()
```

Map data to visual elements or parameters

```
aes(
  x = year,
  y = population,
  color = country
)
```

gg is for Grammar of Graphics

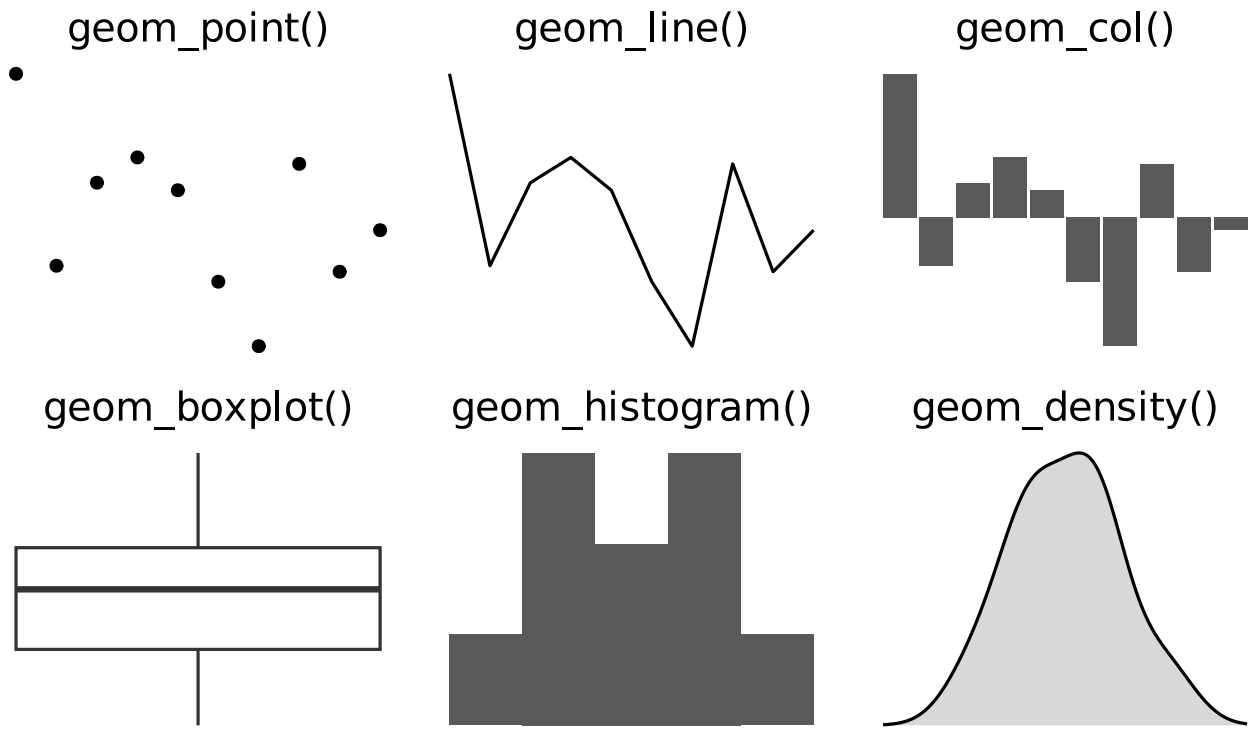
Data

Aesthetics

Geoms

```
+ geom_*()
```

Geometric objects displayed on the plot



gg is for Grammar of Graphics

Data

Aesthetics

Geoms

```
+ geom_*()
```

Here are the some of the most widely used geoms

Type	Function
Point	`geom_point()`
Line	`geom_line()`
Bar	`geom_bar()`, `geom_col()`
Histogram	`geom_histogram()`
Regression	`geom_smooth()`
Boxplot	`geom_boxplot()`
Text	`geom_text()`
Vert./Horiz. Line	`geom_{vh}line()`
Count	`geom_count()`
Density	`geom_density()`

https://eric.netlify.com/2017/08/10/most-popular-ggplot2-geoms/

gg is for Grammar of Graphics

Data

Aesthetics

Geoms

```
+ geom_*()
```

Just start typing `geom_` in `RStudio` to see all the options

```
ggplot(df_geom) +
  aes(x, y) +
  |
```

(Y)Our first plot!

```
gapminder_mean
```

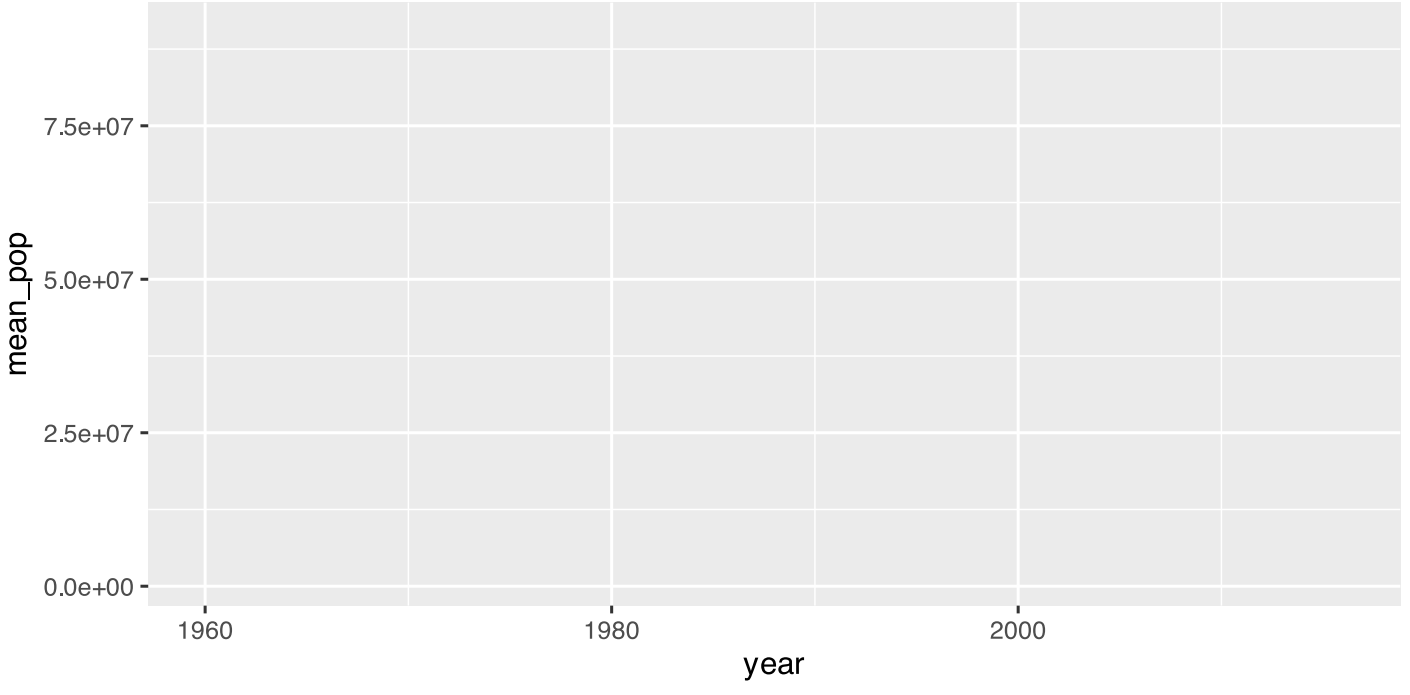
```
## # A tibble: 285 × 3
## # Groups:   continent [5]
##    continent  year mean_pop
##    <fct>     <int>    <dbl>
##  1 Africa     1960 5464985.
##  2 Africa     1961 5598112.
##  3 Africa     1962 5736073.
##  4 Africa     1963 5878867.
##  5 Africa     1964 6026474.
##  6 Africa     1965 6178906.
##  7 Africa     1966 6336258.
##  8 Africa     1967 6498656.
##  9 Africa     1968 6666202.
## 10 Africa     1969 6839011.
## # ℹ 275 more rows
```

(Y)Our first plot!

```
gapminder_mean %>%
  ggplot()
```

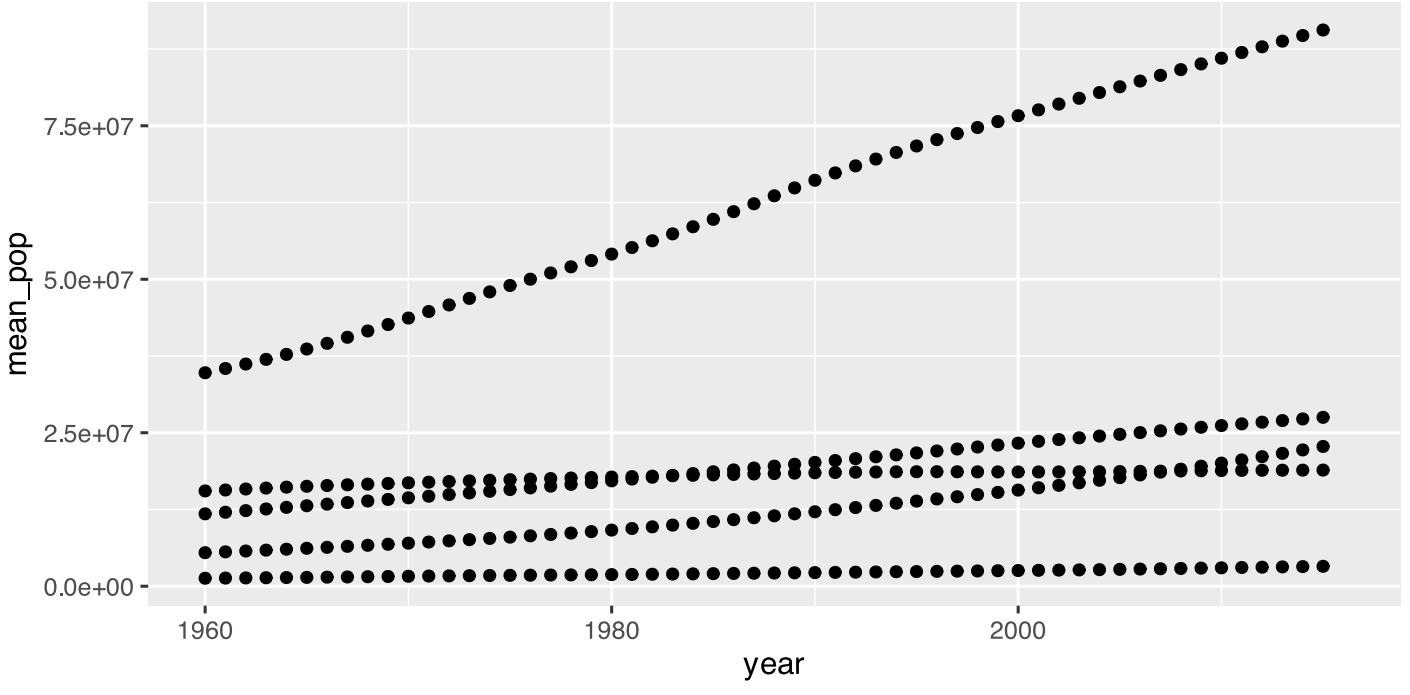
(Y)Our first plot!

```
gapminder_mean %>%
  ggplot() +
  aes(x = year,
      y = mean_pop)
```



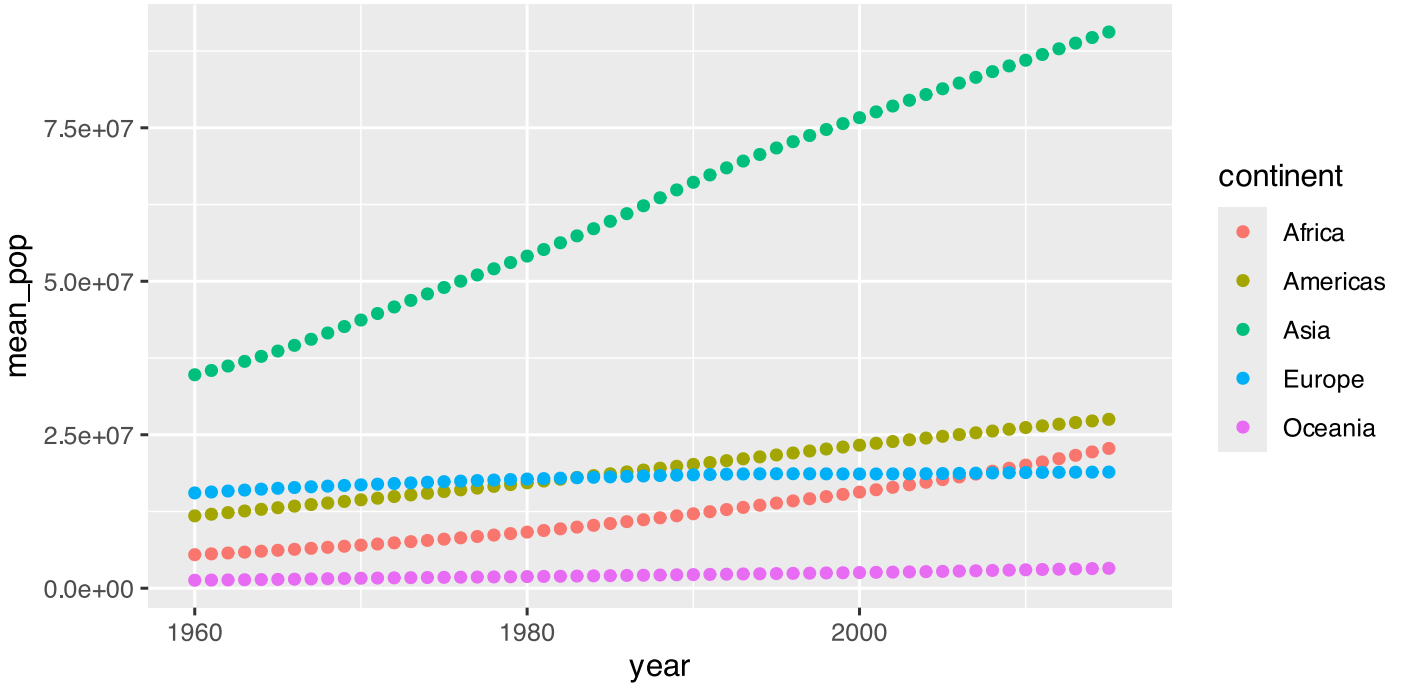
(Y)Our first plot!

```
gapminder_mean %>%
  ggplot() +
  aes(x = year,
      y = mean_pop) +
  geom_point()
```



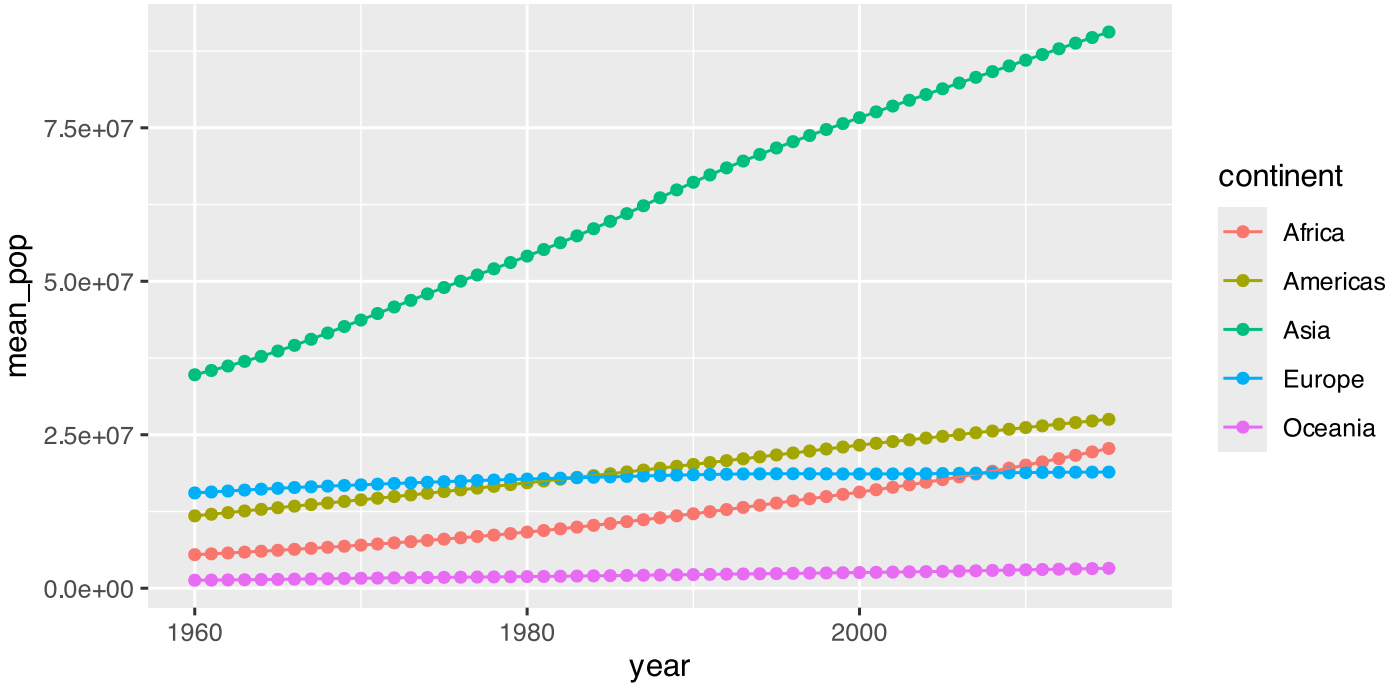
(Y)Our first plot!

```
gapminder_mean %>%
  ggplot() +
  aes(x = year,
      y = mean_pop,
      color = continent) +
  geom_point()
```



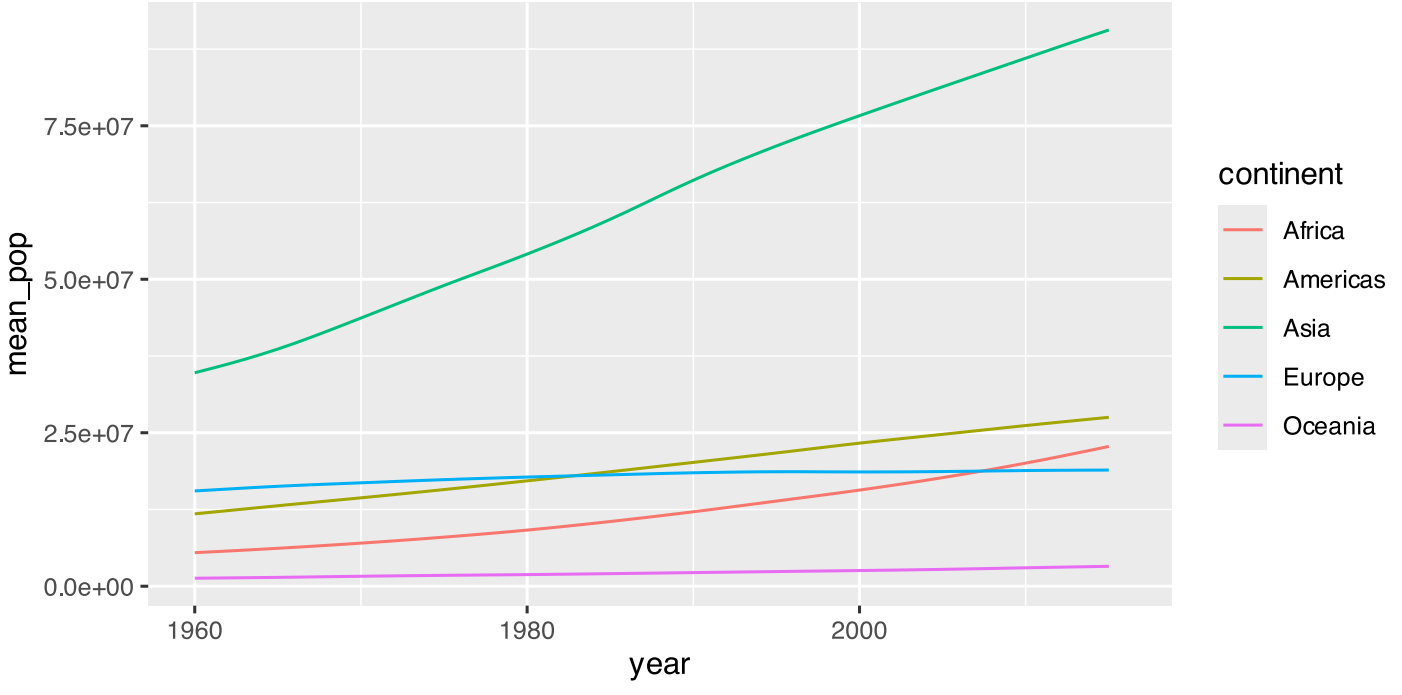
(Y)Our first plot!

```
gapminder_mean %>%
  ggplot() +
  aes(x = year,
      y = mean_pop,
      color = continent) +
  geom_point() +
  geom_line()
```



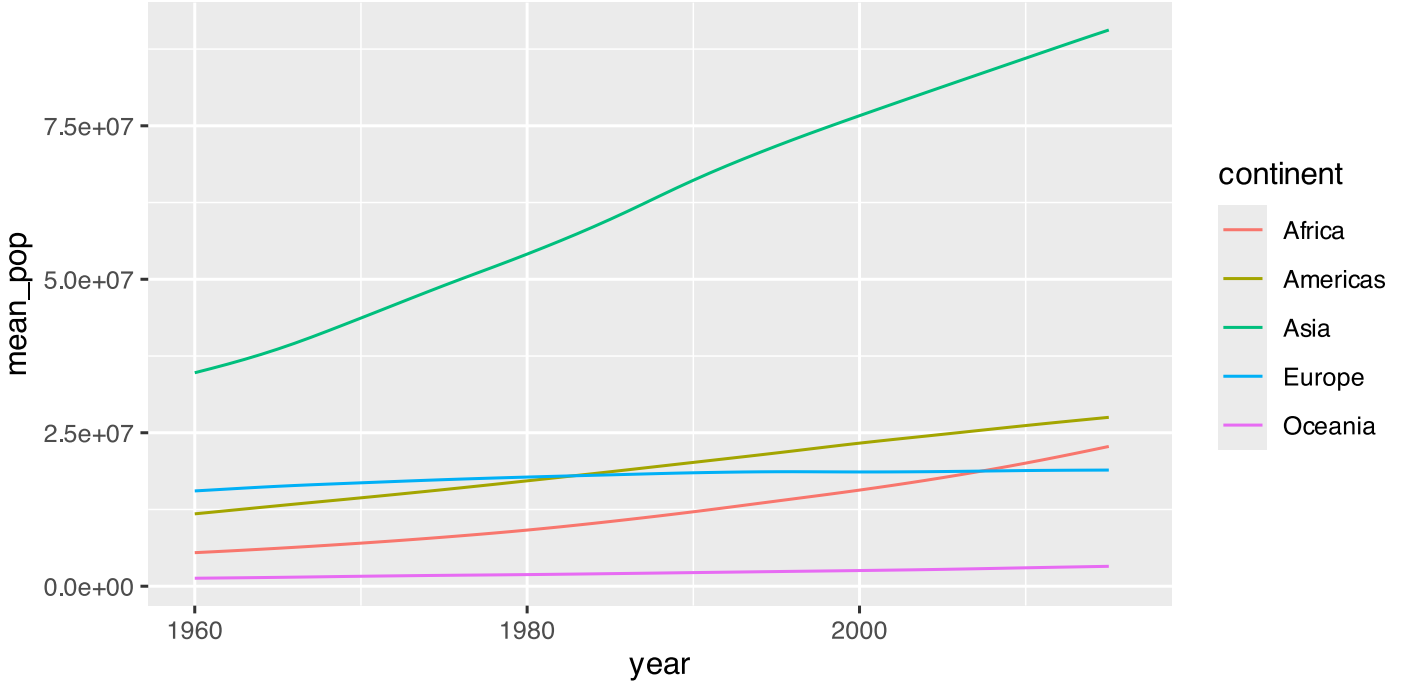
(Y)Our first plot!

```
gapminder_mean %>%
  ggplot() +
  aes(x = year,
      y = mean_pop,
      color = continent) +
  # geom_point() +
  geom_line()
```



(Y)Our first plot!

```
g = gapminder_mean %>%
  ggplot() +
  aes(x = year,
      y = mean_pop,
      color = continent) +
  # geom_point() +
  geom_line()
g
# graphs can be saved as
# objects!
```



gg is for Grammar of Graphics

Data

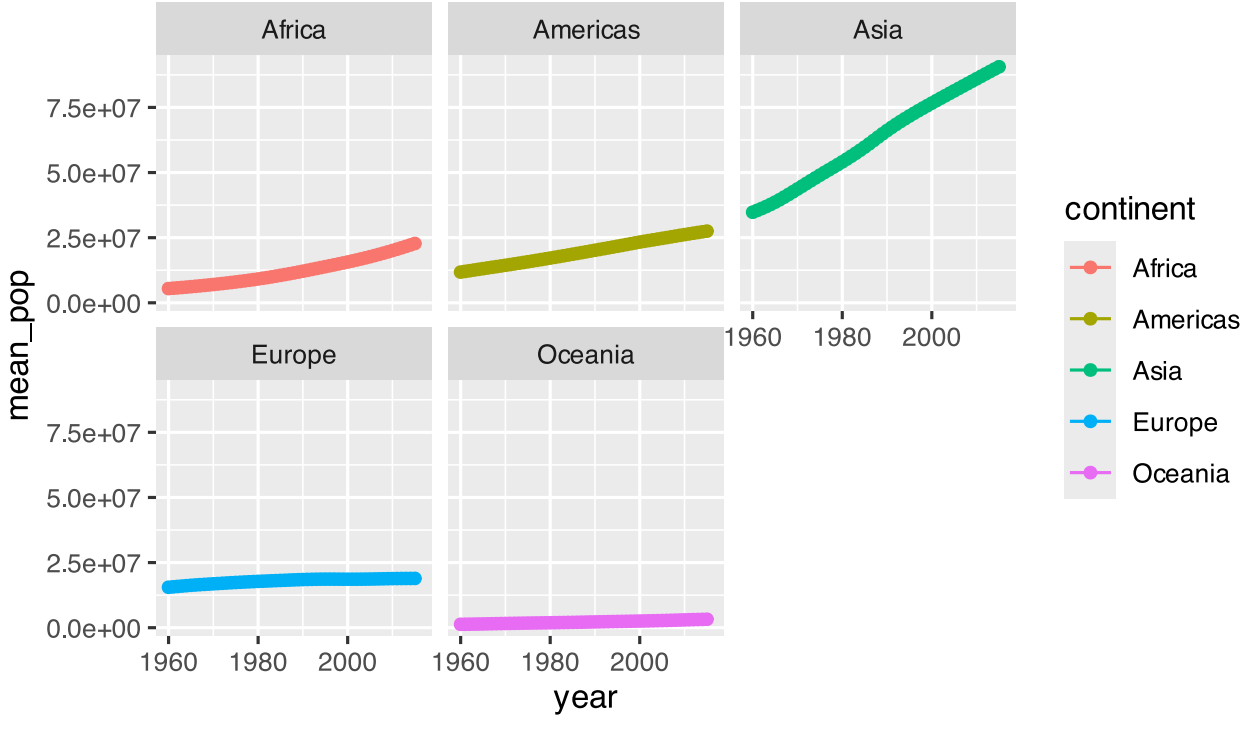
Aesthetics

Geoms

Facet

```
+ facet_wrap()
+ facet_grid()
```

```
g + facet_wrap(~ continent)
```



gg is for Grammar of Graphics

Data

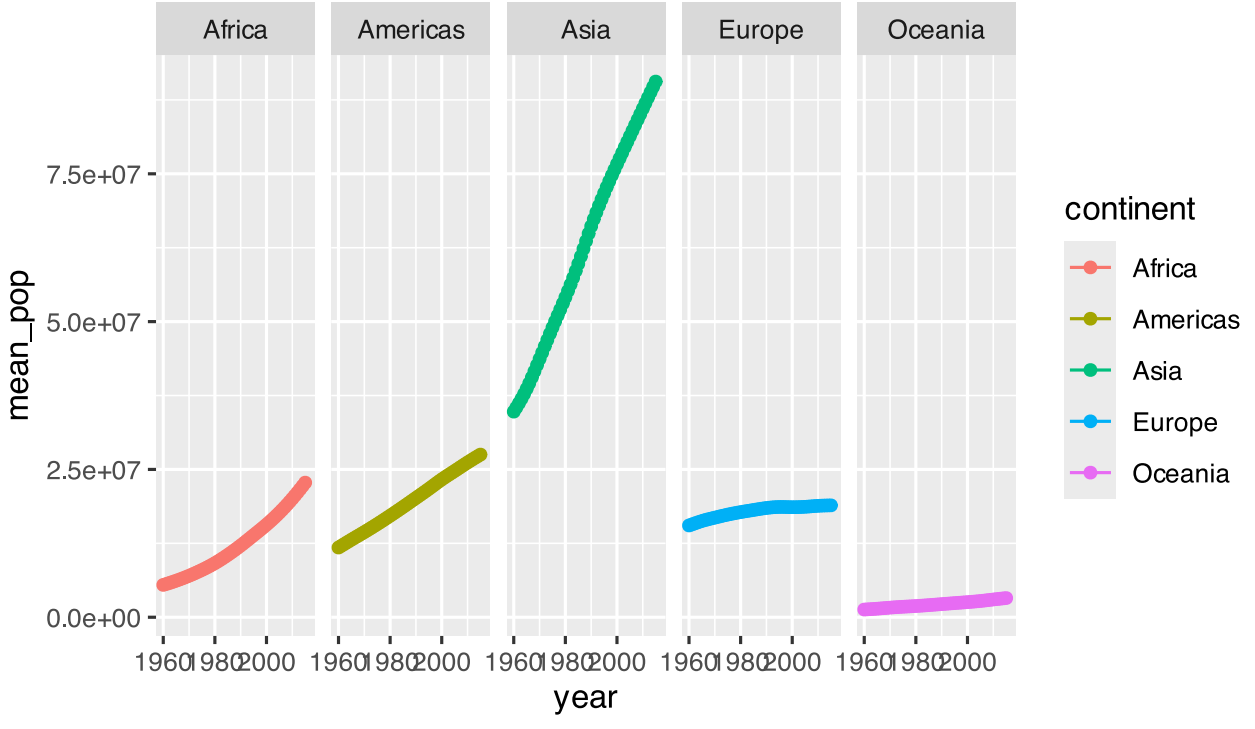
Aesthetics

Geoms

Facet

```
+ facet_wrap()
+ facet_grid()
```

```
g + facet_grid(~ continent)
```



gg is for Grammar of Graphics

Data

Aesthetics

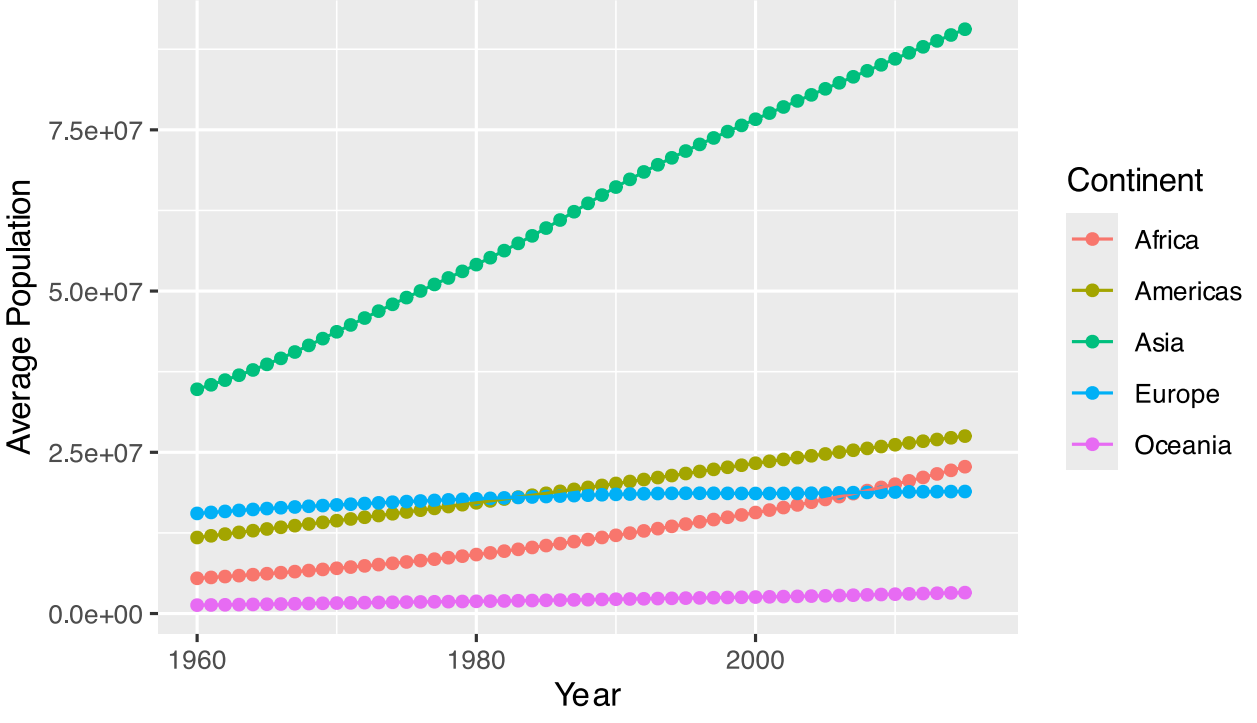
Geoms

Facet

Labels

```
+ labs()
```

```
g + labs(x = "Year", y = "Average Population", color = "Continent")
```



gg is for Grammar of Graphics

Data

Aesthetics

Geoms

Facet

Labels

Scales

```
+ scale_*_*()
```

`scale` + `_` + `<aes>` + `_` + `<type>` + `()`

What parameter do you want to adjust? → `<aes>`
What type is the parameter? → `<type>`

- I want to change my discrete x-axis
  `scale_x_discrete()`
- I want to change range of point sizes from continuous variable
  `scale_size_continuous()`
- I want to rescale y-axis as log10
  `scale_y_log10()`
- I want to use a different color palette
  `scale_fill_discrete()`
  `scale_color_manual()`

gg is for Grammar of Graphics

Data

Aesthetics

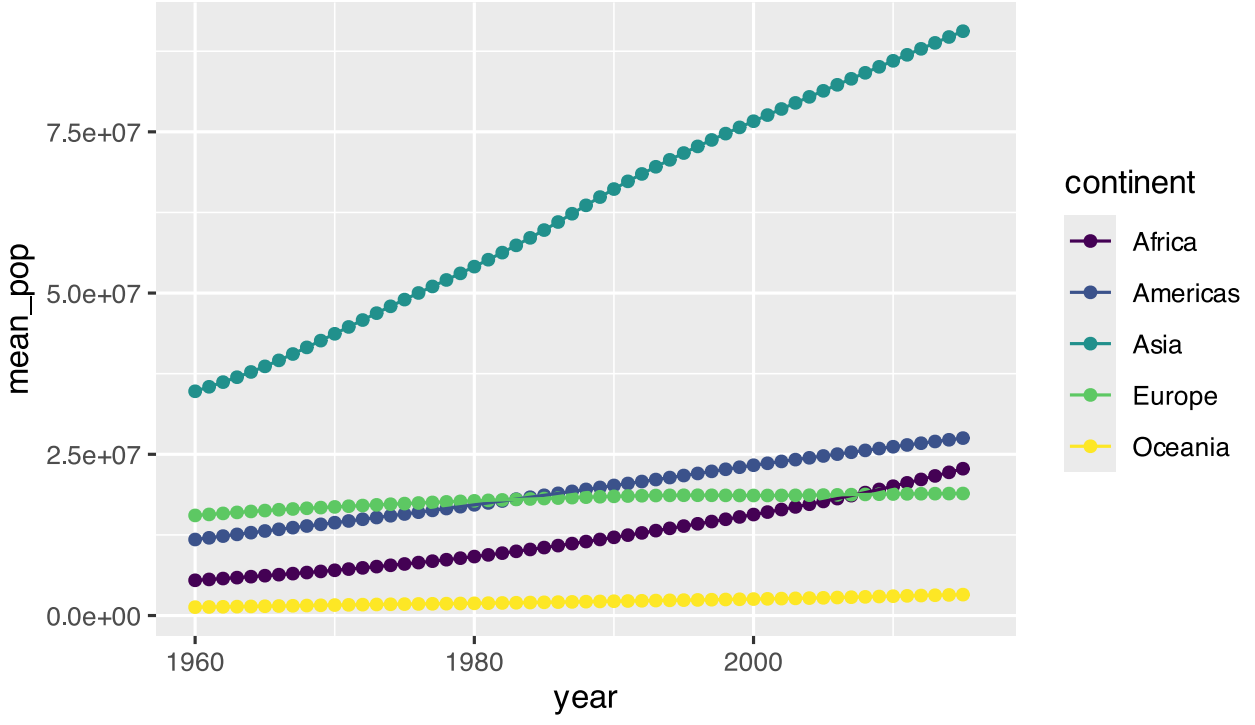
Geoms

Facet

Labels

Scales

```
g + scale_color_viridis_d()
```




```
+ scale_*_*()
```

gg is for Grammar of Graphics

Data

Aesthetics

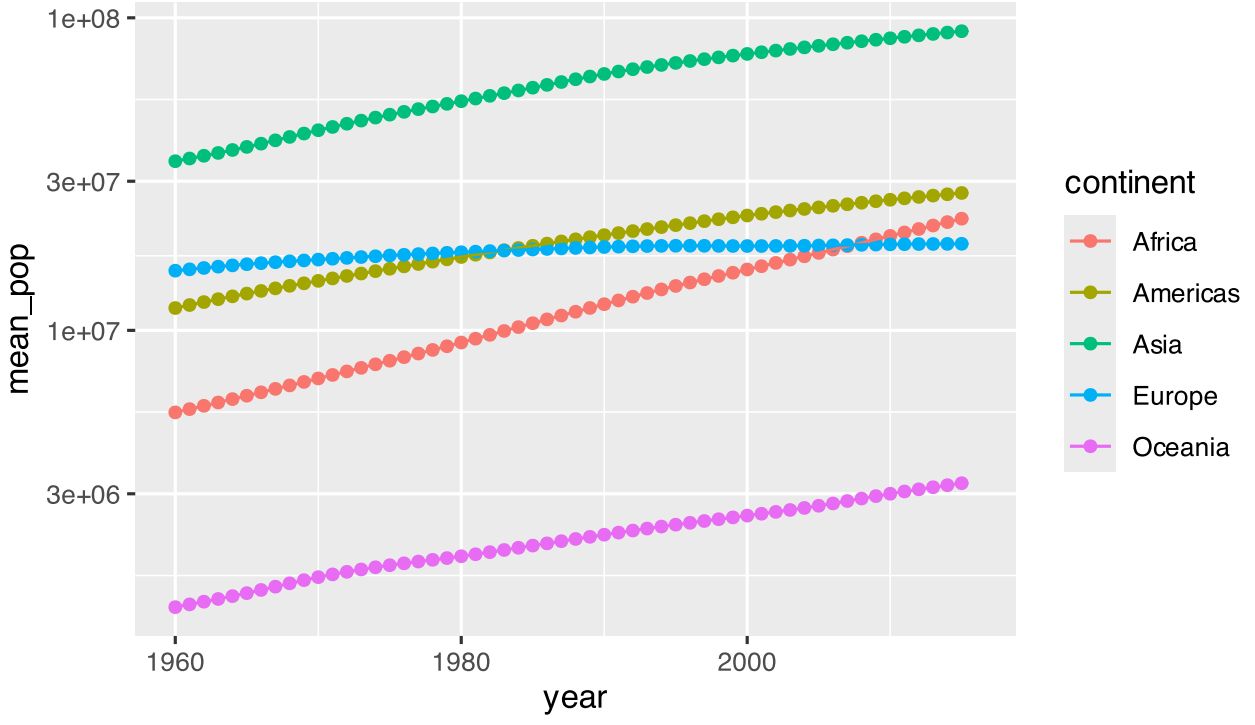
Geoms

Facet

Labels

Scales

```
g + scale_y_log10()
```




```
+ scale_*_*()
```

gg is for Grammar of Graphics

Data

Aesthetics

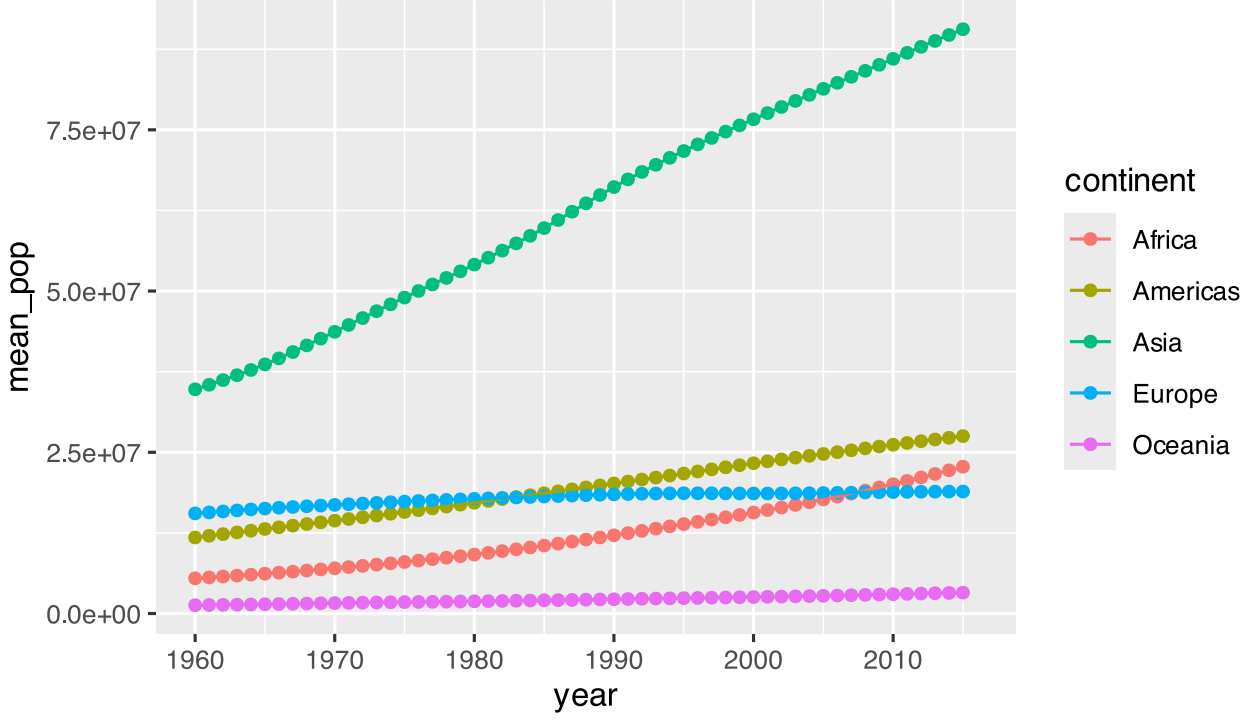
Geoms

Facet

Labels

Scales

```
g + scale_x_continuous(breaks = seq(1950, 2020, 10))
```




```
+ scale_*_*()
```

Delving Deeper into ggplot

- Each graph is different and `ggplot2` provides a zillion options to customize your graph to perfection.

Delving Deeper into ggplot

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- Excellent cheatsheet on project website.

Delving Deeper into ggplot

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Types of Plots

Histograms: counts how many obserations fall within a certain bin.

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Boxplots: displays the distribution of a variable.

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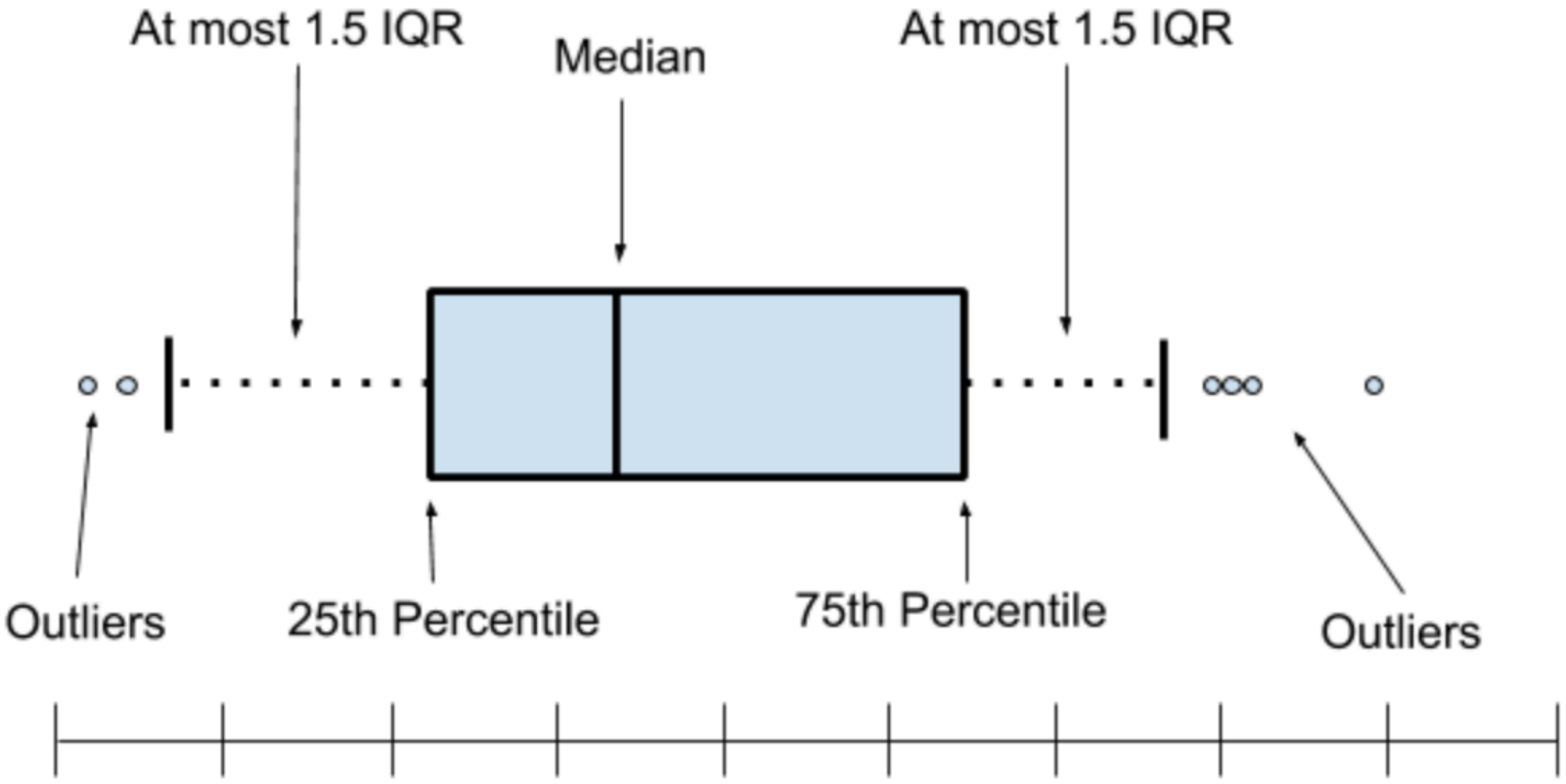

Types of Plots

Histograms: counts how many obserations fall within a certain bin.

Boxplots: displays the distribution of a variable.

Scatter plots: shows the association between two variables.

Task 3: Visualising data

10:00

Using the `gapminder` data, create the following plots using `ggplot2`.

1. A histogram of life expectancy in 2015. (*Hint: do you need to specify a* `y` *in* `aes()` *for a histogram?*) Once you've created the histogram, within the appropriate `geom_*` set: `binwidth` to 5, `boundary` to 45, `colour` to "white" and `fill` to "#d90502". What does each of these options do?
   Optional: Using the previous graph, facet it by continent such that each continent's plot is a new row. (*Hint: check the help for* `facet_grid`.)

2. A boxplot of average life expectancy per year by continent. Within the appropriate `geom_*` set: `colour` to "black" and `fill` to "#d90502". (*Hint: you need to group by both* `continent` *and* `year`.)

3. A scatter plot of fertility rate (y-axis) with respect to infant mortality (x-axis) in 2015. Once you've created the scatter plot, within the appropriate `geom_*` set: `size` to 3, `alpha` to 0.5, `colour` to "#d90502". Add labels (`labs`) to the plot so that it is cleaner.

Summarising

missing

joint and marginal dists

1. conditional dists, independence, covariance, corr
2. LIE
3. correlation and conditional mean

create/get a table and illustrate

Summarising Data

- In general, we can learn from the data by visualising it and/or computing summary statistics

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- In general, we can learn from the data by visualising it and/or computing summary statistics
- Let's now turn to summary statistics!

Summarising Data

- In general, we can learn from the data by visualising it and/or computing summary statistics
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- In order to **summarize** properly, we need to step back and remind ourselves of what ***Probability*** and ***Distributions*** are.

Summarising Data

- In general, we can learn from the data by visualising it and/or computing summary statistics
- Let's now turn to summary statistics!
- In order to **summarize** properly, we need to step back and remind ourselves of what ***Probability*** and ***Distributions*** are.
- (All of this will of course be recap from your stats classes. 🧐)

UNIVERSITÀ DI TORINO
Department of
Economics, Social Studies,
Applied Mathematics
and Statistics

Enough `tidy`ing! On to Probability!

	Slides
	My Homepage
	Book